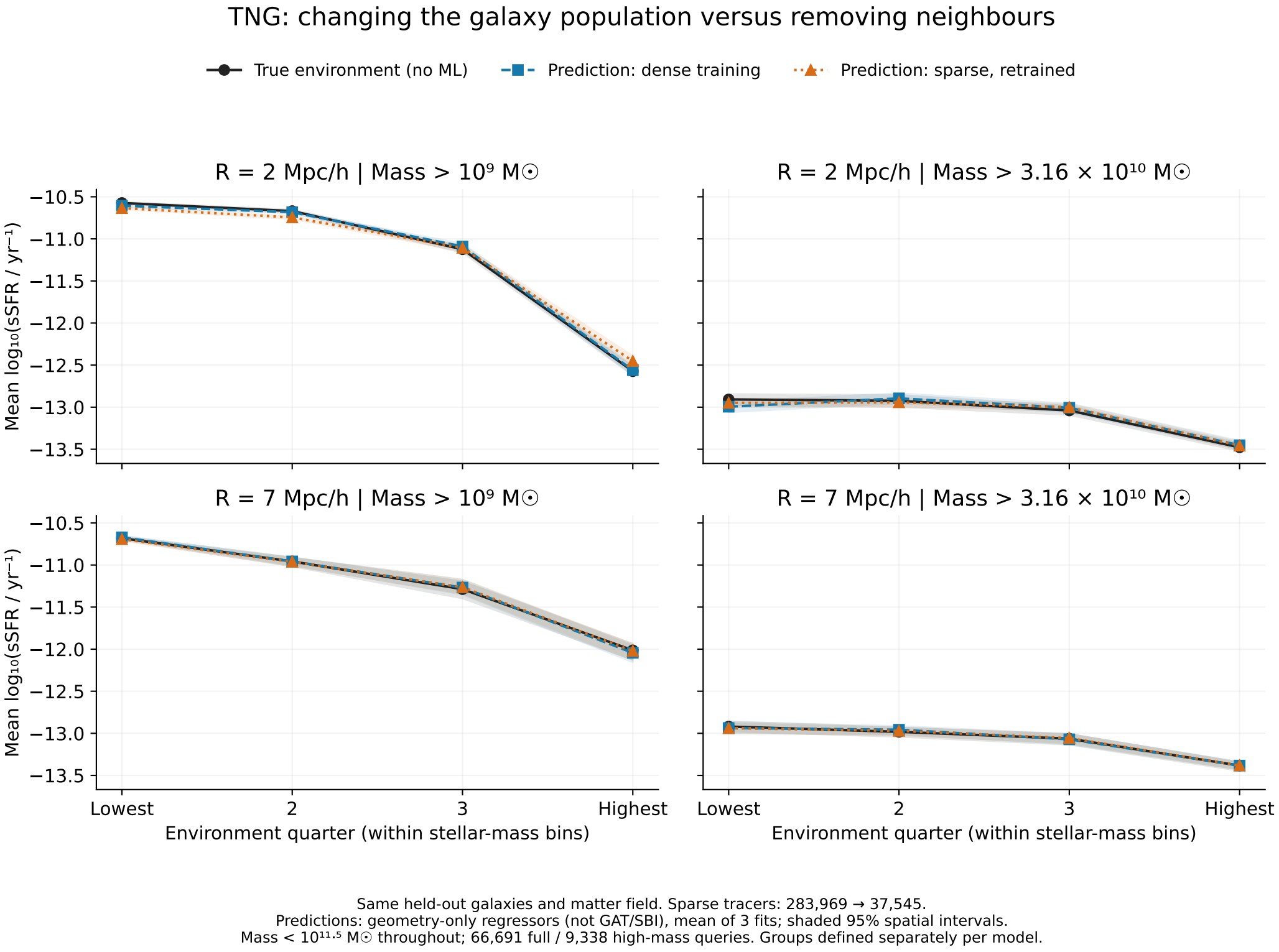
TNG: changing the galaxy population versus removing neighbours
True environment (no ML)
Prediction: dense training
Prediction: sparse, retrained
R = 2 Mpc/h | Mass > 10⁹ M⊙
R = 2 Mpc/h | Mass > 3.16 × 10¹⁰ M⊙
R = 7 Mpc/h | Mass > 10⁹ M⊙
R = 7 Mpc/h | Mass > 3.16 × 10¹⁰ M⊙
Mean log₁₀(sSFR / yr⁻¹)
−10.5
−11.0
−11.5
−12.0
−12.5
−13.0
−13.5
Lowest
2
3
Highest
Environment quarter (within stellar-mass bins)
Same held-out galaxies and matter field. Sparse tracers: 283,969 → 37,545.
Predictions: geometry-only regressors (not GAT/SBI), mean of 3 fits; shaded 95% spatial intervals.
Mass < 10¹¹·⁵ M⊙ throughout; 66,691 full / 9,338 high-mass queries. Groups defined separately per model.